



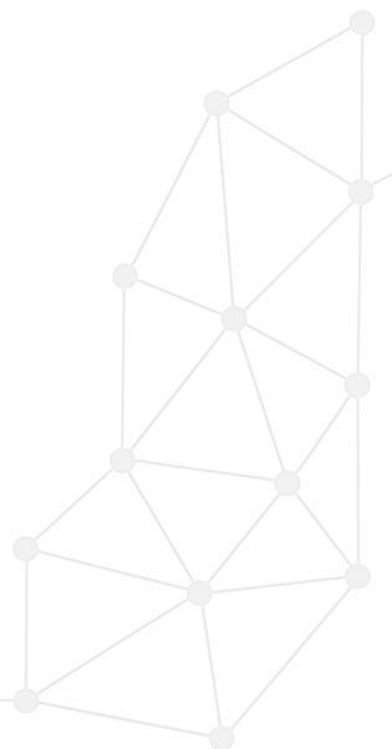
PROGETTO FINALE MODULO 2

“File e Directory pt.1” a cura di FABIO RENNA



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Executive Summary

Nel presente lavoro viene documentato il processo di installazione, utilizzo e analisi del gioco **GameShell** all'interno di un ambiente Kali Linux, con l'obiettivo di favorire la familiarizzazione con i principali comandi della shell e con la navigazione del filesystem Linux.

Si è proceduto all'installazione del software seguendo la procedura indicata nella traccia, per poi eseguire un insieme di livelli rappresentativi. Per ogni livello è stata effettuata la lettura della missione, la navigazione tra directory e file necessari per il completamento, e la validazione finale tramite i comandi forniti dalla piattaforma ("**gsh goal**" e "**gsh check**").

La presente relazione sintetizza gli obiettivi del progetto, le attività svolte, la logica delle missioni, i principali comandi utilizzati e le evidenze ottenute durante l'avanzamento di livello, presentando una guida operativa strutturata e replicabile.

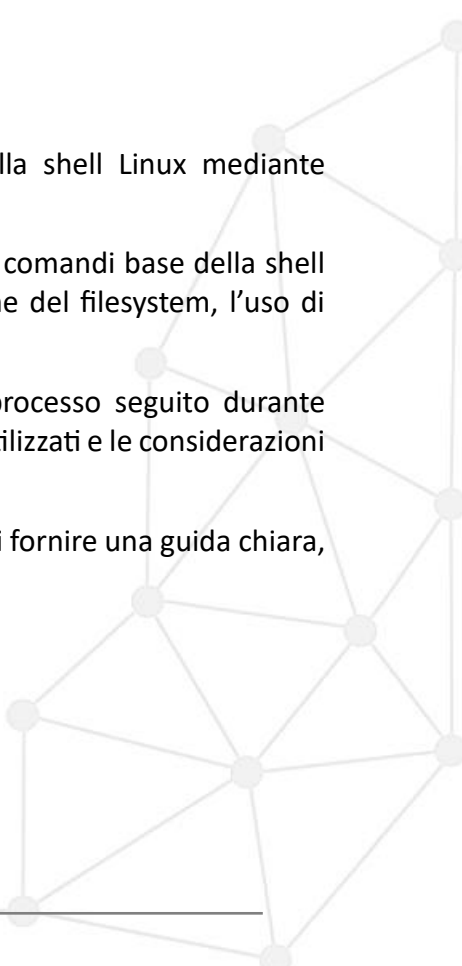
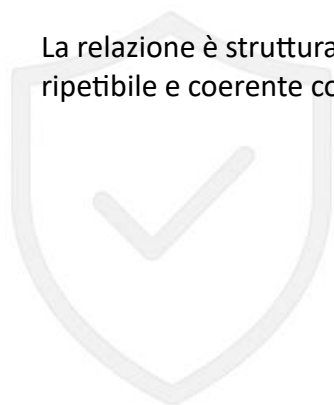
Introduzione

Il progetto si propone di introdurre lo studente all'utilizzo pratico della shell Linux mediante l'installazione e l'esecuzione del gioco **GameShell** su ambiente Kali Linux.

GameShell è un'applicazione progettata per favorire l'apprendimento dei comandi base della shell attraverso una serie di missioni progressive che richiedono la navigazione del filesystem, l'uso di comandi fondamentali e il raggiungimento di specifici obiettivi operativi.

Nel contesto del presente modulo, viene richiesto di documentare il processo seguito durante l'esecuzione dei livelli, evidenziando la logica di risoluzione, gli strumenti utilizzati e le considerazioni tecniche emerse.

La relazione è strutturata in modo formale e impersonale, con l'obiettivo di fornire una guida chiara, ripetibile e coerente con le indicazioni del docente.



Successivamente si è proceduto a lanciare il gioco e ad iniziare il completamento delle missioni

```
Session Actions Edit View Help

GameShell

Press Enter to continue.
```

```
Session Actions Edit View Help

Welcome to GameShell!

Using and administering a computer running a Unix-like operating system (e.g., Linux, BSD or macOS) requires using a command line interface called "shell". In this game you will learn how to use the shell
by Journeying through various missions that will teach you to perform standard tasks such as:
- creating directories,
- creating, viewing and editing files,
- searching for files satisfying certain criteria,
- running and interrupting programs,
- etc.

During your adventure you will need to visit locations, interact with people, and find various objects. However, you have to keep in mind that these are standard directories, programs, and files (often
with random contents). So, for example:
- "build a hut" means creating a directory called "hut", and
- "put a coin in your chest" means moving a file called "coin" to a directory called "Chest".

Press Enter to continue.
```

```
Session Actions Edit View Help

Run the command
$ gsh goal
to discover your first mission.

You can check the mission has been completed with
$ gsh check

The command
$ gsh help
displays the list of available (gsh) commands.

[mission 1] $
```

La missione numero 1 prevede l'uso dei comandi **cd** ed **ls** per muoverci all'interno del castello per raggiungere il tetto della torre:

```
Session Actions Edit View Help

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Castle Forest Garden Mountain Stall

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Castle

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Cellar Great_hall Main_building Main_tower Observatory

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Main_tower

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
First_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd First_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Second_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Second_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Top_of_the_tower

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Top_of_the_tower

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls

[use 'gsh help' to get a list of available commands]
[mission 1] $ gsh check

Congratulations, mission 1 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]
```

La missione 2 vuole che l'utente si muova nella cartella "Cellar" tramite i comandi **pwd** (mostra il percorso corrente), **cd ..** (torna indietro nel percorso di 1 passo) e **cd -** (torna indietro nel percorso precedente di 1 passo) capendo le rispettive funzionalità:

```
Session Actions Edit View Help

[use 'gsh help' to get a list of available commands]
[mission 2] $ ls

[use 'gsh help' to get a list of available commands]
[mission 2] $ pwd
/home/kali/gameshell/World/Castle/Main_tower/First_floor/Second_floor/Top_of_the_tower

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd -
/home/kali/gameshell/World/Castle/Main_tower/First_floor/Second_floor

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd ..

[use 'gsh help' to get a list of available commands]
[mission 2] $ ls
Second_floor

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd -
/home/kali/gameshell/World/Castle/Main_tower/First_floor/Second_floor

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd -
/home/kali/gameshell/World/Castle/Main_tower/First_floor

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd -
/home/kali/gameshell/World/Castle/Main_tower/First_floor/Second_floor

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd ..

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd ..

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd ..

[use 'gsh help' to get a list of available commands]
[mission 2] $ ls
Cellar Great_hall Main_building Main_tower Observatory

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd Cellar

[use 'gsh help' to get a list of available commands]
[mission 2] $ gsh check

Congratulations, mission 2 has been successfully completed!
```


La missione 4 chiede di costruire un “chest” all’interno di un “Hut” nella Foresta. Si è utilizzato il comando **mkdir** per creare “Hut” e “Chest” come da richiesta.

```
[mission 4] $ gsh goal

Mission goal
=====

Build a "Hut" in the forest, and then build a "Chest" in the hut.

Useful commands
=====

mkdir DIRECTORY
Create a new directory inside the current directory.
Remark: ``mkdir`` is an abbreviation for "make directory".

~
[mission 4] $ cd Forest
~/Forest
[mission 4] $ mkdir Hut
~/Forest
[mission 4] $ cd Hut
~/Forest/Hut
[mission 4] $ mkdir Chest
~/Forest/Hut
[mission 4] $ gsh check

Congratulations, mission 4 has been successfully completed!
```

La missione 5 chiede di eliminare tutti i files contenenti la parola “spiders” all’interno della Cellar e di lasciare i “bats” da soli. Quindi si è proceduto ad eliminare con il comando **rm**.

```
[mission 5] $ gsh goal

Mission goal
=====

Go back to the cellar and get rid of all the spiders. Leave the bats alone: they appear on the castle's coat of arms and are said to confer luck.

Useful commands
=====

rm FILE1 FILE2 ... FILEn
Delete the files (permanently).
Remark: ``rm`` is an abbreviation for "remove".

~
[mission 5] $ cd Castle/Cellar
~/Castle/Cellar
[mission 5] $ ls
barrel_of_apples  bat_1  bat_2  spider_1  spider_2  spider_3
~/Castle/Cellar
[mission 5] $ rm -r spider_1 | rm -r spider_2 | rm -r spider_3
~/Castle/Cellar
[mission 5] $ ls
barrel_of_apples  bat_1  bat_2
~/Castle/Cellar
[mission 5] $ rm -r barrel_of_apples
~/Castle/Cellar
[mission 5] $ gsh check

Congratulations, mission 5 has been successfully completed!
```

La missione 6 chiede di spostare i “coins” dal giardino al nostro “chest”, utilizzando il comando **mv**.

```
kali@kali: ~
Session Actions Edit View Help
[mission 6] $ gsh goal

Mission goal
Collect all the coins that you can find in the garden in front of the castle, and put them in your chest in your hut in the forest.

Useful commands
mv FILE1 FILE2 ... FILEn DIRECTORY
Move the files to the directory.
Remark: `mv` is an abbreviation of "move".

The "~" symbol is an abbreviation for the initial directory.
Example: wherever you are, `~/Tavern` denotes the directory (or file) "Tavern" in the initial directory.

[mission 6] $ cd Garden
~/Garden
[mission 6] $ mv coin_1 ~/Forest/Hut/Chest | mv coin_2 ~/Forest/Hut/Chest | mv coin_3 ~/Forest/Hut/Chest
~/Garden
[mission 6] $ gsh check
Congratulations, mission 6 has been successfully completed!
```

La missione 7 è simile alla prima, ma chiede di spostare i “coins” nascosti, che andranno listati con il comando **ls -A** e poi spostarli con **mv**.

```
()=(
Mission goal
Collect all the coins hidden in the garden in front of the castle, and put them in your chest (in your hut in the forest).

Secondary objective
Learn how to use the "Tab" key to go faster.

Useful commands
ls -A
List all the files of the current directory, including hidden files. (A file is "hidden" when its name starts with a dot.)
Tab
The tabulation key "completes" the name of a file or directory once you have typed the beginning of its name. This only works
if there is only one possible completion.
Tab-Tab
Pressing tabulation twice successively shows a list of possible completions.

()=(
~/Garden
[mission 7] $ ls -A
.27915_coin_2 .33549_coin_1 .49706_coin_3 Flower_garden Maze Shed
~/Garden
[mission 7] $ mv .27915_coin_2 ~/Forest/Hut/Chest | mv .33549_coin_1 ~/Forest/Hut/Chest | mv .49706_coin_3 ~/Forest/Hut/Chest
~/Garden
[mission 7] $ gsh check
Congratulations, mission 7 has been successfully completed!
```

La missione 8 chiede di eliminare i files “spiders” lasciando gli altri files. Per fare ciò si è utilizzato il comando **rm** e gli asterischi ***** all’inizio e alla fine della parola spider, indicando di eliminare solo i files che contengono quella parola.

```
Mission goal
Get rid of all the spiders that are crawling in the cellar. Again, do not do not disturb the bats.

Shell patterns
*
The "*" character stands in for any sequence of characters
(including an empty sequence).
?
The "?" character stands in for any single character.
Those wildcards can be used to denote lists of existing files / directories in the current working directory.
For example: if the current folder contains
file-1 Folder-1 file-14 potato
then
*      → file-1 Folder-1 file-14 potato
*1     → file-1 Folder-1
*ov    → Folder-1 potato
**     → error, no matching file
*-?    → file-1 Folder-1
*-??   → file-14

()=(

~/Castle/Cellar
[mission 8] $ cd ~/Castle/Cellar
~/Castle/Cellar
[mission 8] $ ls -la
1112_spider_46 12530_spider_17 15184_spider_5 1742_bat_4 23579_spider_31 27193_spider_27 28904_spider_40 30752_spider_18 32093_spider_49 32504_spider_43 3720_spider_41 5977_spider_6 6940_spider_14 9195_spider_16
12079_spider_37 12772_bat_5 16040_spider_3 19065_spider_23 24255_spider_29 27666_spider_32 29456_spider_10 30885_spider_15 32327_spider_36 3333_spider_30 3962_spider_21 6257_spider_2 7189_spider_47 9251_spider_34
12230_spider_9 13201_spider_45 16792_spider_42 20849_spider_19 26847_spider_4 27989_spider_11 29974_spider_12 31211_spider_1 32356_spider_20 3484_spider_35 4479_spider_26 6483_spider_28 7677_bat_3 9780_spider_7
12389_spider_25 13908_spider_8 17103_spider_22 23328_spider_44 27124_bat_1 28233_spider_30 30052_spider_50 31247_spider_48 3239_spider_39 3689_spider_13 5806_spider_33 6729_bat_2 8593_spider_24

~/Castle/Cellar
[mission 8] $ rm spiders*
~/Castle/Cellar
[mission 8] $ gsh check
Congratulations, mission 8 has been successfully completed!
```

La missione 9 è simile alla precedente e chiede di eliminare i files nascosti contenuti “spiders”. Quindi, prima di **rm *spider*** si fa una lista di file nascosti utilizzando il comando **ls -A**.

```
()=(

Mission goal
The spiders are getting clever: they found a way to hide.
Get rid of all the spiders that are hiding in the cellar without disturbing the bats.

Shell patterns
*
The "*" character stands in for any sequence of characters (including an empty sequence).
?
The "?" character stands in for any single character.

Remark
The wildcards "*" and "?" don't see hidden files, you need to add an explicit dot at the start of the pattern.

()=(

~/Castle/Cellar
[mission 9] $ ls -A
.1039_spider_4 12629_spider_26 .147_spider_37 .16431_spider_23 .18421_spider_10 .19946_spider_22 .23816_spider_48 27124_bat_1 .29314_spider_20 .31919_spider_7 .5195_spider_21 7677_bat_3
.10650_spider_45 12772_bat_5 .15316_spider_19 .16620_spider_16 .19024_spider_3 .20380_spider_1 .24210_bat_4 .27161_spider_24 .30102_spider_2 .32136_spider_38 .6069_spider_30 .8653_spider_17
.10881_spider_28 .13165_spider_46 .15518_spider_47 .16741_spider_15 .19194_spider_34 .20433_spider_50 .24372_spider_29 .27335_spider_36 .30167_spider_39 .3472_spider_43 6729_bat_2 .8837_spider_42
.11070_spider_6 .13265_spider_5 .15550_bat_5 .17023_spider_25 .19620_spider_18 .21088_spider_33 .2590_spider_49 .27375_spider_14 .30461_spider_32 .3984_spider_27 .7206_spider_13 .9052_spider_9
.1188_bat_1 .14494_spider_31 .15611_bat_3 1742_bat_4 .1963_bat_2 .22220_spider_12 .26452_spider_4 .28550_spider_44 .3103_spider_11 .5169_spider_40 .7429_spider_35 .9415_spider_41

~/Castle/Cellar
[mission 9] $ rm .spider*
~/Castle/Cellar
[mission 9] $ gsh check
Congratulations, mission 9 has been successfully completed!
```

La missione 10 chiede di fare una copia utilizzando il comando **cp** per rubare i files standard nel nostro "Chest".

```
~/Castle
[mission 10] $ gsh goal

Mission goal
-----
You have taken a fancy to the four standards in the great hall of the castle. As stealing them would not go unnoticed, put a copy (same name, same content) of each in your chest.

Useful commands
-----
cp FILE DIRNAME
Copy the file to the directory.
Remark: `cp` is an abbreviation of "copy".

~/Castle
[mission 10] $ cd Great_hall

~/Castle/Great_hall
[mission 10] $ ls
25153_decorative_shield 40892_stag_head 47089_suit_of_armour standard_1 standard_2 standard_3 standard_4

~/Castle/Great_hall
[mission 10] $ cp standard_1 standard_2 standard_3 standard_4 ~/Forest/Hut/Chest

~/Castle/Great_hall
[mission 10] $ gsh check

Congratulations, mission 10 has been successfully completed!
```

La missione 11 chiede di copiare i files contenenti la parola "tapestry" nel "Chest". Per fare ciò sono stati utilizzati i comandi **cp** seguiti dagli asterischi ***** per copiare tutti i files contemporaneamente.

```

The tapestries in the castle's great hall are also particularly beautiful. Put a copy of each in your chest.

Useful commands
-----
cp FILE1 FILE2 ... FILEn DIRNAME
Copy the files to the directory.
Remark: `cp` is an abbreviation of "copy".

Shell patterns
-----
*
The "*" character stands in for any sequence of characters
(including an empty sequence).
?
The "?" character stands in for any single character.

~/Castle/Great_hall
[mission 11] $ ls
16863_tapestry_09 17718_tapestry_06 33805_tapestry_03 40519_tapestry_01 49788_tapestry_05 54072_tapestry_04 57846_tapestry_10 standard_2 standard_4
17361_tapestry_02 24093_tapestry_08 37092_suit_of_armour 43856_tapestry_07 4978_decorative_shield 55922_stag_head standard_1 standard_3

~/Castle/Great_hall
[mission 11] $ cp *tapestry* ~/Forest/Hut/Chest

~/Castle/Great_hall
[mission 11] $ gsh check

Congratulations, mission 11 has been successfully completed!
```

La missione 12 chiede di copiare nel nostro “Chest” il file più vecchio tra quelli elencati. Per fare ciò si è utilizzato il comando `ls -l`, mostrando la data di ultima modifica. Una volta individuato, si è spostato il file dove richiesto.

```
~
[mission 12] $ gsh goal

Mission goal
-----
While wandering around the first floor of the main tower, some magnificent paintings catch your eye. Add a copy of the oldest one to your chest.

Secondary objectives
-----
Take a moment to admire the sheer beauty of the paintings.

Useful commands
-----
ls -l
  Print the list of files of the current directory, with additional information including last modification date.

cat FILE
  Display the contents of the file.

~
[mission 12] $ cd ~/Castle/Main_tower/First_floor
~/Castle/Main_tower/First_floor
[mission 12] $ ls -l
total 16
-rw-rw-r-- 1 kali kali 1055 May 13 1986 painting_bCtVngGm
-rw-rw-r-- 1 kali kali 1455 Aug 18 2014 painting_bQLCqykC
-rw-rw-r-- 1 kali kali 1503 Aug 20 1995 painting_geUZqrCF
drwxrwxr-x 3 kali kali 4096 Nov 13 13:26 Second_floor/
~/Castle/Main_tower/First_floor
[mission 12] $ cp painting_bCtVngGm ~/Forest/Hut/Chest
~/Castle/Main_tower/First_floor
[mission 12] $ gsh check

Congratulations, mission 12 has been successfully completed!
```



La missione 13 chiede di individuare il giorno all'interno di una data [02-21-2028].

Utilizzando il comando `cal <year>` si può avere un print dell'intero calendario dell'anno inserito.

Una volta individuato il giorno esatto, si è inserito nella risposta per confermare la missione.

```
~/Castle/Main_tower/First_floor
[mission 13] $ gsh goal

Mission goal

Nostradamus predicted a spectacular star conjunction on the 02-21-2028.
But what will the day of the week be on that date?

When you have it, run the command ``gsh check``.

Useful commands

cal
Print a calendar for the current month.

cal YEAR
Print a calendar for the given year.

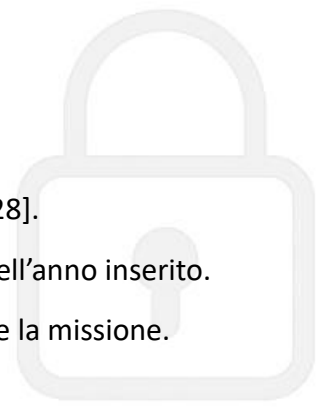
This mission is optionnal. You can skip it and go to the next one with the
command

$ gsh skip

~/Castle/Main_tower/First_floor
[mission 13] $ cal 2028
      2028
January February March
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
26 27 28 29 30 31 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29 30 31

~/Castle/Main_tower/First_floor
[mission 13] $ gsh check
What was the day of the week for the 02-21-2028?
1 : Monday
2 : Tuesday
3 : Wednesday
4 : Thursday
5 : Friday
6 : Saturday
7 : Sunday
Your answer: 1

Congratulations, mission 13 has been successfully completed!
```



La missione 14 chiede semplicemente di creare una scorciatoia per il comando `ls -A`.

Per fare ciò basta utilizzare il comando **alias**, nel caso specifico si è utilizzata l'abbreviazione **la** per indicare **ls -A**.

```

Mission goal

Checking for hidden files is taking too long!

Create an alias "la" to run the command ``ls -A`` in order to list all files, including hidden ones, with only 2 letters.

Define the synonym

la

for the command

ls -A

and check that it works as expected.

How fortunate, there is a nice rock hidden just where you are.

Useful commands

alias STRING='COMMAND'
Create a synonym for a string, that will stand for a command.

~/Castle/Main_tower/First_floor
[mission 14] $ alias la='ls -A'

~/Castle/Main_tower/First_floor
[mission 14] $ la
.nice_rock painting_bCtVngGm painting_bQLCqyk painting_geUZqrCF Second_floor/

~/Castle/Main_tower/First_floor
[mission 14] $ gsh check

Congratulations, mission 14 has been successfully completed!
```

La missione 15 chiede semplicemente di creare un nuovo file di testo "journal.txt" all'interno di "Chest".

Per completare la missione si è utilizzato l'editor di testo, avviandolo con il comando **nano**.

```

~/Castle/Main_tower/First_floor
[mission 15] $ gsh goal

Mission goal

Create a file named "journal.txt" in your chest and write a short message in it.
You can use this file to record your notes and solutions for the upcoming missions.

Details

``nano`` is a command-line text editor. You can use it whenever you need to edit a file from the shell.

Useful commands

nano FILE
Edit the file from the shell.
(If the file does not exist, it will be created.)

Keybindings are listed at the bottom of the screen (the ``^`` symbol means "Control"). The most important ones are:
Control-x quit
Control-o save
Control-w search for a string

Remark: do not use Control-s or Control-z!

~/Castle/Main_tower/First_floor
[mission 15] $ cd ~/Forest/Hut/Chest

~/Forest/Hut/Chest
[mission 15] $ nano

~/Forest/Hut/Chest
[mission 15] $ ls
16863_tapestry_09 17718_tapestry_06 33805_tapestry_03 43856_tapestry_07 54872_tapestry_04 coin_1 coin_3 painting_bCtVngGm standard_2 standard_4
17361_tapestry_02 24093_tapestry_08 40519_tapestry_01 49789_tapestry_05 57846_tapestry_10 coin_2 journal.txt standard_1 standard_3

~/Forest/Hut/Chest
[mission 15] $ gsh check

Congratulations, mission 15 has been successfully completed!
```



La missione 16 chiede di creare un alias come fatto precedentemente per il comando **ls -A**, ma questa volta per il file journal.txt. Si è utilizzato lo stesso comando precedente: **alias journal='journal.txt'**.

```
~/Forest/Hut/Chest
[mission 16] $ gsh goal

/b\~\
(  \~\~\
Mission goal
||
|| Create an alias "journal" in order to easily edit your journal file wherever you are.
||
||
|| Details
||
|| To edit the journal file with ``nano`` from, for example, the cellar or the throne room, you need to give the full path to the file: "~/Forest/.../journal.txt".
||
|| To avoid typing this long command each time, you can create an alias just like
||
||     alias la='ls -a'
||
|| Useful commands
||
|| nano FILE
||     Edit the file from the shell.
||     (If the file does not exist, it will be created.)
||
|| alias STRING='COMMAND'
||     Create a synonym for a string, that will stand for a command.
||
|| Note
||
|| If (and only if) you know what you're doing, you can use the ``EDITOR`` variable to define your favorite editor and use it in the alias.
||
/b\~\
(  \~\~\

~/Forest/Hut/Chest
[mission 16] $ alias journal='nano ~/Forest/Hut/Chest/journal.txt'

~/Forest/Hut/Chest
[mission 16] $ gsh check

Congratulations, mission 16 has been successfully completed!
```



La missione17 chiede di eliminare entro 20 secondi il file contenente la parola "spider queen". Per velocizzare il processo si è utilizzato il comando **TAB** che restituisce il nome completo di un file o directory all'interno del percorso dopo aver inserito semplicemente la lettera iniziale.

```
(0)=>>.....=<=(0)
)
) Mission goal
)
) At the back of the cellar, there is a small opening going to the spider queen's lair.
) Go there, and remove the spider queen (and nothing else).
)
) Note: you have a limited amount of time (20 seconds) to do that. You can use the command ``gsh reset`` to reset the timer.
) Another thing: shell patterns have been deactivated. You cannot use the wildcards ``*`` or ``?``.
)
) Useful commands
)
) Tab
) The "Tabulation" key completes the name of a file or directory once you have typed the beginning of its name. This only works
) if there is only one possible completion.
)
) Tab-Tab
) Pressing the "Tabulation" key twice successively shows a list of possible completions.
)
).....=<=(0)
)
~/Castle/Cellar
[mission 17] $ gsh reset

|-----|
| Use the command |
| $ gsh help      |
| to get the list of "gsh" commands. |
|-----|

~/Castle/Cellar
[mission 17] $ cd .Lair_of_the_spider_queen\ LavmYvaPxCcyGo0z LbtPSckuRllLWKi0/
~/Castle/Cellar/.Lair_of_the_spider_queen LavmYvaPxCcyGo0z LbtPSckuRllLWKi0
[mission 17] $ ls
AjpGjPKmTddJfdZL_baby_bat_yMavQedelHwRCJB PekllsSt0iQsKEVA_spider_queen_uRLHZmToiEbRwSsS
~/Castle/Cellar/.Lair_of_the_spider_queen LavmYvaPxCcyGo0z LbtPSckuRllLWKi0
[mission 17] $ rm PekllsSt0iQsKEVA_spider_queen_uRLHZmToiEbRwSsS
```

```
~/Castle/Cellar/.Lair_of_the_spider_queen LavmYvaPxCcyGo0z LbtPSckuRllLWKi0
[mission 17] $ gsh check
Perfect, it took you only 16 seconds to complete this mission!

Congratulations, mission 17 has been successfully completed!
```



La missione 18 chiede di eseguire un processo/comando chiamato **xeyes** e fermarlo. Successivamente, eseguirlo di nuovo per poi metterlo in background utilizzando il comando **&**.

```
~/Castle/Cellar/.Lair_of_the_spider_queen LavmYvaPxCcyGo0z LbtPSCkuRllLWKi0
[mission 18] $ gsh goal

Mission goal
As you are walking around the castle, you feel like you are being watched... Turn your head quickly enough and you may see one of the paintings' eyes following you.
1/ Run the ``xeyes`` command, and stop it.
2/ Run the ``xeyes`` command in the background.

Useful commands
xeyes
  Open a window with 2 eyes that track your mouse.
COMMAND &
  Run the command in the background.
Control-c (also written ^c)
  Pressing Control and c at the same times interrupts the current command by sending the INT ("INTerrupt") signal to the process.

This mission is optionnal. You can skip it and go to the next one with the
command
$ gsh skip

~/Castle/Cellar/.Lair_of_the_spider_queen LavmYvaPxCcyGo0z LbtPSCkuRllLWKi0
[mission 18] $ xeyes

This mission is optionnal. You can skip it and go to the next one with the
command
$ gsh skip

~/Castle/Cellar/.Lair_of_the_spider_queen LavmYvaPxCcyGo0z LbtPSCkuRllLWKi0
[mission 18] $ xeyes
xeyes &
^C

~/Castle/Cellar/.Lair_of_the_spider_queen LavmYvaPxCcyGo0z LbtPSCkuRllLWKi0
[mission 18] $ xeyes &
[1] 41156

~/Castle/Cellar/.Lair_of_the_spider_queen LavmYvaPxCcyGo0z LbtPSCkuRllLWKi0
[mission 18] $ gsh check

Congratulations, mission 18 has been successfully completed!
[ progress was saved in /home/kali/gameshell-save.sh ]

Use the command
$ gsh help
to get the list of "gsh" commands.
```

La missione 19 chiede di utilizzare il comando **flarigo** per far apparire dei fuochi d'artificio e metterlo in background in modo da tenerlo in esecuzione quando si fa il check del goal.

```
~/Castle/Cellar/.Lair_of_the_spider_queen LavmYvaPxCcyGo0z LbtPSCkuRllLWKiO
[mission 19] $ gsh goal

(0) >===== (0)

Mission goal

The king's pyrotechnician appears next to you. He asks you to fire **at least 3 consecutive fireworks** so he can see them from far away.

A single firework can be created with the magical word

    flarigo

Useful commands

flarigo
    This (non standard) command creates a single small firework.

COMMAND &
    Run the given command, but don't wait until it is finished to return.
    The command will run in the "background".

COMMAND1 ; COMMAND2 ; ... ; COMMANDn
    Run the given commands one after the other.
    Each command is run when the previous one is finished.

COMMAND1 & COMMAND2 & ... & COMMANDn
    Run the given commands "in parallel".
    All the commands are run in the "background", except the last one.

(0) >===== (0)

~/Castle/Cellar/.Lair_of_the_spider_queen LavmYvaPxCcyGo0z LbtPSCkuRllLWKiO
[mission 19] $ gsh check & flarigo & flarigo & flarigo
[2] 42482
[3] 42483
[4] 42485
*
*
*

Let's have a look:
```

```

      *
    *
  *
* ' ' *
* _ \ _ *
* ^ *
* .. *

      *
      :
    '.\./.'
    -= o -=
    './.\.'
      :
      *
    .:::.
    .:::.
    .:::

[3]- Done          flarigo
[4]+ Done          flarigo

~/Castle/Cellar/.Lair_of_the_spider_queen LavmYvaPxCcyGo0z LbtPSCkuRllLWKiO
[mission 19] $ Great, that looked good!

Congratulations, mission 19 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]

|
+-----+
| Use the command |
|   $ gsh help   |
| to get the list of "gsh" commands. |
+-----+
|
```


La missione 21 chiede di trovare un file che contenesse la parola “copper coin” all’interno di un labirinto di directories e files. Per questa missione si è proceduto semplicemente con **ls** e **cd** per trovare il file corretto.

```
00000_copper_coin_00000
~/Garden/Maze/7204f1887564be4bec7430da320834b/4cf16325cdf85e10b2c8d07424bc/ae16352f6a91783232d796083697
[mission 21] $ mv <copper> ~/Forest/Hut/Chest
~/Garden/Maze/7204f1887564be4bec7430da320834b/4cf16325cdf85e10b2c8d07424bc/ae16352f6a91783232d796083697
[mission 21] $ ls
~/Garden/Maze/7204f1887564be4bec7430da320834b/4cf16325cdf85e10b2c8d07424bc/ae16352f6a91783232d796083697
[mission 21] $ cd ~/Forest/Hut/Chest
~/Forest/Hut/Chest
[mission 21] $ ls
16863_tapestry_09 17718_tapestry_06 33805_tapestry_03 43856_tapestry_07 54072_tapestry_04 coin_1 coin_3 00000_copper_coin_00000 painting_bCtVngGm standard_2 standard_4
17361_tapestry_02 24093_tapestry_08 40519_tapestry_01 49788_tapestry_05 57846_tapestry_10 coin_2 00000_copper_coin_00000 standard_1 standard_3
~/Forest/Hut/Chest
[mission 21] $ gsh check
Congratulations, mission 21 has been successfully completed!
```

La missione 22, invece, chiede di fare la stessa cosa ma con un file che contenesse la parola “silver coin” e con una moltitudine di directories e files. Procedere con **ls** e **cd** risulta molto complicato, indi per cui è stato utilizzato il comando **tree**, che mostra un albero di files e directories, partendo dalla posizione corrente.

```
~/Forest/Hut/Chest
[mission 22] $ gsh goal

Mission goal

Find the silver coin in the maze in the garden and move it to your chest using the shell.

Useful commands

ls -R
Print the list of all files / directory, including those in sub-directories (recursively).

tree
Print the tree of files and directories, starting from the current working directory.
```

```
09a850110c6e53a34810ef496ec51
104241d172574bb3d134bdfadfb
├── 36723b211a2e36638d7a8
├── 45fe9f94b49892f1
├── c02e5ed52176030d570b
├── 26bf2fde9a91
├── 9f407e015a
├── da2bfa7c3b5de
├── da3699dd4fcd5
├── 8f0c385748037b331207dd33233e023
├── 6ee4044e6e3f1fa259f6d4ab852
├── 7329f11975b07ea96ff48fbad1
├── 8deb2685b919a69
├── 09a8dfddcd1d5d
├── 8bdec3ac8352ace9d3ef5
├── 7127ffa92ba8cbf4bce
├── 859e4561
├── ba3f32b6aff31c83dad
├── 975f4613d1
├── 6472b28e64eaea7fc
├── d9c2540b7ce7d119
├── eb03bfa4ce12bf
├── f43892bbcd
├── 09cfbe879ae37a0618bf9db132f8ed
├── 2f6a873e75bdeb567a
├── e7f674f4cc8edfb2dfb6fda61
├── a656be377ee49e0836
├── 0a3c30bc5ad40b4e8
├── 328f0612c7b5
├── 41ed99a3b5ae58e5ec60e9101fe1ab5
├── 5a634f9d010e49190950743
├── 231a9277313a26d9b1
├── 17b48816
├── 8a4e2208627eae31dbcfab8ce1d
├── ab9c1d36c73825fba27a7
├── 85543095371c6bc1e
├── 05a8e7abaf20
├── 2d12adfbce8
├── 5ba729535157cf5e20cb326ee3
└── 00000_silver_coin_00000
40 directories, 1 file
~/Garden/Maze
[mission 22] $ mv ~/Garden/Maze/a656be377ee49e0836/85543095371c6bc1e/5ba729535157cf5e20cb326ee3/00000_silver_coin_00000 ~/Forest/Hut/Chest/
~/Garden/Maze
[mission 22] $ gsh check
Congratulations, mission 22 has been successfully completed!
```

La missione 23 complica ancor di più l'esercizio, chiedendo di trovare dei "gold coin" all'interno di migliaia di directories e spostarli nel "Chest". Utilizzare il comando **tree** sarebbe inefficace, quindi, si è utilizzato il comando **find -iname** (la "i" ignora le maiuscole e minuscole) seguito dalla parola chiave tra asterischi ***** e/o apici **"** per trovare rapidamente i percorsi dei files e permettere il completamento della missione.

```
~/Garden/Maze
[mission 23] $ mv ~/Garden/Maze/e215155ffed/fd6d7db5bd049c955c3f0b7db5/ef121d22ec68a704db48379dea31/gold_coin_1 ~/Forest/Hut/Chest/

~/Garden/Maze
[mission 23] $ find -name *gold*

~/Garden/Maze
[mission 23] $ find -name *coin*

~/Garden/Maze
[mission 23] $ find -iname "gold"

~/Garden/Maze
[mission 23] $ find -iname "g"

~/Garden/Maze
[mission 23] $ find -iname "G"

~/Garden/Maze
[mission 23] $ find -iname *g*
./b123677027267e5f/79ecc91d1a/b888d18fcc0/GolD_CoiN_2

~/Garden/Maze
[mission 23] $ find -name ?gold

~/Garden/Maze
[mission 23] $ find -name ?gold?

~/Garden/Maze
[mission 23] $ mv ~/Garden/Maze/b123677027267e5f/79ecc91d1a/b888d18fcc0/GolD_CoiN_2 ~/Forest/Hut/Chest/

~/Garden/Maze
[mission 23] $ goal check
Command 'goal' not found, did you mean:
  command 'gol' from deb growl-for-linux
  command 'gcal' from deb gcal
  command 'gdal' from deb gdal-bin
Try: sudo apt install <deb name>

~/Garden/Maze
[mission 23] $ gsh check

Congratulations, mission 23 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]
```

La missione 24 chiede di leggere un numero limitato all'interno di un file di testo. Per fare questo, si è utilizzato il comando **head -n 6**. Head indica che il conteggio delle stringhe che si vogliono stampare parta dall'alto, mentre -n 6 indica il numero di stringhe che si vogliono stampare. In questo caso verranno lette solo le prime 6 stringhe del file page_07.

```
~/Mountain/Cave
[mission 24] $ gg

Mission goal
-----
A forgetful old hermit called Servillus has set up camp in a cave with his old, leather-bound potion book.
Go to the cave and help him remember the recipe of his famous herbal tea.
In order to validate the mission, you need to be in the cave with Servillus **and** your last command prior to "gsh check" must show the recipe (including its title), but nothing else.
Note: you shouldn't alter the content of the book of potions.

Useful commands
-----
cat FILE
  Display the contents of the file.
head FILE
  Print the first 10 lines of the file.
head -n K FILE
  Print the first K lines of the file.

Remark
-----
A "FILE" may contain directories if the file in question is not in the current directory.

~/Mountain/Cave
[mission 24] $ head -n 6 ~/Mountain/Cave/Book_of_potions/page_07
~~~~~
Herbal tea
~~~~~
1) Boil water.
2) Add herbs from the forest.
3) Let it sit for five minutes and drink while hot.

~/Mountain/Cave
[mission 24] $ gsh check

Congratulations, mission 24 has been successfully completed!
```

La missione 25 chiede, invece di leggere un numero di stringhe alla fine di un documento di testo. Per fare ciò, a differenza del comando head che parte dall'alto, si è utilizzato il comando **tail -n 9**, che appunto indica quante stringhe stampare (in questo caso 9) partendo dall'ultima del documento. (page_12)

```
~/Mountain/Cave
[mission 25] $ gg

Mission goal

The old man seems to enjoy your company very much. He invites you to stay for supper, and starts preparing a delicious stew for the both of you. While getting the cauldron ready he asks for your help.
Read him the steps of the recipe from his book.

In order to validate the mission, you need to be in the cave with Servillus **and** last command prior to ``gsh check`` must show the steps of the recipe (without its title).

Note: you shouldn't alter the content of the book of potions.

Useful commands

cat FILE
  Display the contents of the file.

tail FILE
  Print the last 10 lines of the file.

tail -n K FILE
  Print the last K lines of the file.

~/Mountain/Cave
[mission 25] $ tail -n 9 ~/Mountain/Cave/Book_of_potions/page_12
1) Boil water in a cauldron.
2) Add in a few death caps (Amanita phalloides).
3) Also add a few fly agarics (Amanita muscaria).
4) And some destroying angels (Amanita virosa).
5) Mix in a few deadly webcaps (Cortinarius rubellus).
6) Feel free to add in any colourful fungi you have on hand.
7) Let half of the water evaporate.
8) Season with a pinch of salt and a few herbs.
9) Serve hot in a bowl.

~/Mountain/Cave
[mission 25] $ gsh check
Congratulations, mission 25 has been successfully completed!
```

La missione 26 chiede questa volta di stampare due files consecutivamente, dato che uno è il completamento dell'altro, utilizzando quindi 2 volte il comando **cat**.

```
( While cleaning the dishes, Servillus mentions an interesting potion that lets the drinker (temporarily) take the physical appearance of anyone.
  Read the recipe of the potion from the hermit's book.
  In order to validate the mission, you need to be in the cave with Servillus **and** your last command prior to ``gsh check`` must show the whole recipe (with its title).
  Note: you shouldn't alter the content of the book of potions.
  Useful commands
  cat FILE1 FILE2 ... FILEn
    Display the contents of the files in order.
    Remark: ``cat`` is an abbreviation for "concatenate".
  )

(0)=>=====<=(0)

~/Mountain/Cave
[mission 26] $ cat ~/Mountain/Cave/Book_of_potions/page_01 cat ~/Mountain/Cave/Book_of_potions/page_02
Transformation potion
1) Boil water in a cauldron.
2) Add 3 measures of fluxweed to the cauldron.
3) Add 2 bundles of knotgrass to the cauldron.
4) Stir 4 times, clockwise.
5) Wave your wand then let potion brew for 80 minutes.
6) Add 4 leeches to the cauldron.
7) Crush 2 scoops of lacewing flies to a fine paste.
8) Add 2 measures of the crushed lacewings to the cauldron.
9) Heat for 30 seconds on a low heat.
cat: cat: No such file or directory
10) Add 3 measures of boomslang skin to the cauldron.
11) Crush a bicorn horn into a fine powder.
12) Add 1 measure of the crushed horn to the cauldron.
13) Heat for 20 seconds at a high temperature.
14) Wave your wand then let potion brew for 24 hours.
15) Add 1 additional scoop of lacewings to the cauldron.
16) Stir 3 times, counter-clockwise.
17) Split potion into multiple doses, if desired.
18) Add a pieces of the person you wish to become.
19) Wave your wand to complete the potion.

~/Mountain/Cave
[mission 26] $ gsh check
cat: cat: No such file or directory
Congratulations, mission 26 has been successfully completed!
```

La missione 27 chiede di stampare una lista di stringhe che si completano in 2 documenti di testo differenti eliminando le prime 4. Per completare la missione, oltre ad utilizzare 2 volte il comando **cat** per stampare il documento, si è aggiunto il comando **tail -n 16** per soddisfare la richiesta, stampando solo le ultime 16 stringhe dei documenti.

```
[mission 27] $ cat Book_of_potions/page_03 cat Book_of_potions/page_04
vvvvvvvvvvvvvvvvvv
Elixir of youth
AAAAAAAAAAAAAAAAAAAA
WARNING: **do not** drink if you are older than 30.
1) Fill a cauldron with used bath water.
2) Put a moderately large frog in the water.
3) Let the preparation rest overnight.
4) The next morning thank and free your little green friend.
5) Boil the water and add in a few sticks of oak tree.
6) Crush 5 river stones to a fine powder.
7) Mix in a third of the powder and stir vigorously.
cat: cat: No such file or directory
8) Let the preparation rest for a day.
9) Add hairs from the tail of a squirrel (willingly given).
10) Add the remaining stone powder.
11) Stir the potion very vigorously, in all directions.
12) Take some time to rest after such an effort.
13) Rest a little bit more.
14) Even take a nap if you want.
15) Add a few larch tree needles for seasoning.
16) Drink the potion from the cauldron.

-/Mountain/Cave
[mission 27] $ cat Book_of_potions/page_03 cat Book_of_potions/page_04 | tail -n 16
cat: cat: No such file or directory
1) Fill a cauldron with used bath water.
2) Put a moderately large frog in the water.
3) Let the preparation rest overnight.
4) The next morning thank and free your little green friend.
5) Boil the water and add in a few sticks of oak tree.
6) Crush 5 river stones to a fine powder.
7) Mix in a third of the powder and stir vigorously.
8) Let the preparation rest for a day.
9) Add hairs from the tail of a squirrel (willingly given).
10) Add the remaining stone powder.
11) Stir the potion very vigorously, in all directions.
12) Take some time to rest after such an effort.
13) Rest a little bit more.
14) Even take a nap if you want.
15) Add a few larch tree needles for seasoning.
16) Drink the potion from the cauldron.

-/Mountain/Cave
[mission 27] $ gsh check
cat: cat: No such file or directory

Congratulations, mission 27 has been successfully completed!
```

La missione 28 chiede di stampare le stringhe centrali di un documento. Per fare ciò si è prima utilizzato il comando **head -n 6** e successivamente **tail -n 3**. Unendo i due comandi in questo modo, si è ottenuta la stampa di stringhe centrali di un documento.

```
Mission goal

The old hermit is thirsty and he would like you lookup the recipe for distilled water.

In order to validate the mission, you need to be in the cave with Servillus **and** your last command prior to ``gsh check`` must show the steps for the recipe and nothing else.

Note: you shouldn't alter the content of the book of potions.

Useful commands

head [-n K] [FILE]
Print the first lines (10 by default, K if ``-n K`` is used) of the given file, or standard input if no file is given.

tail [-n K] [FILE]
Print the last lines (10 by default, K if ``-n K`` is used) of the given file, or standard input if no file is given.

COMMAND1 | COMMAND2
Run the two commands, feeding the "standard output" of the former into the "standard input" of the latter.
Remark: by analogy with plumbing "I" is called "pipe".

Remark

Parts in [square brackets] in command descriptions are stand for optional parts. The brackets are not part of the command.

~/Mountain/Cave
[mission 28] $ head -n 6 Book_of_potions/page_13 | tail -n 3
1) Boil water in a big pot.
2) Condense the vapor in a fresh container.
3) Add minerals for a better taste (optional).

~/Mountain/Cave
[mission 28] $ gsh check
Congratulations, mission 28 has been successfully completed!
```

La missione 29 chiede semplicemente di fermare un processo in esecuzione. Per completare si è prima utilizzato il comando **ps** per vedere quali fossero i processi in esecuzione e una volta individuato, è stato utilizzato il comando **kill**, seguito dal PID del processo stesso.

```
~/Mountain/Cave
[mission 29] $ kill 72145
bash: kill: (72145) - No such process

~/Mountain/Cave
[mission 29] $
  *#@*
  8_**/~
  !$-#

  *#@*
  8_**/~
  !$-#

  *#@*
  8_**/~
  !$-#

  *#@*
  8_**/~
  !$-#

^C
~/Mountain/Cave
[mission 29] $ kill 72415

~/Mountain/Cave
[mission 29] $ gsh check
Congratulations, mission 29 has been successfully completed!
```

La missione 30 chiede di fare esattamente la stessa cosa, ma con un processo che è impossibile da terminare utilizzando solo il comando **kill**. Per completare la missione è stata aggiunta un'opzione **kill -9** che forza la chiusura di un processo in esecuzione.

```
~/Mountain/Cave
[mission 30] $ ps
  PID TTY          TIME CMD
 49087 pts/0    00:00:00 zsh
 49122 pts/0    00:00:00 bash
 49162 pts/0    00:00:01 bash
 73653 pts/0    00:00:00 spell
 74676 pts/0    00:00:00 ps

~/Mountain/Cave
[mission 30] $
    *#@*
    6_**/~
    !$-#

    *#@*
    6_**/~
    !$-#

    *#@*
    6_**/~
    !$-#

^C

~/Mountain/Cave
[mission 30] $ kill -9 73653

~/Mountain/Cave
[mission 30] $ gsh check

Congratulations, mission 30 has been successfully completed!
```



La missione 31 richiede di eliminare solo gli *spell* eseguiti da un processo malevolo, in questo caso ***mischievous_imp***, lasciando intatti tutti gli altri. Poiché i comandi `ps` o `pstree` da soli non sono sufficienti per individuare i PID dei sottoprocessi appartenenti a un processo specifico, è stato utilizzato il comando **`pstree -p`** seguito dal PID del “*mischievous_imp*”. In questo modo si possono visualizzare sia i nomi dei sottoprocessi sia i loro PID, facilitando la loro identificazione e la successiva terminazione selettiva.

```
~/Castle/Cellar
[mission 31] $ pstree -p 175284
mischievous_imp(175284)─spell(175310)─sleep(207957)
                        └─spell(175311)─sleep(208029)
                        └─spell(175312)─sleep(207886)
                        └─tail(175313)

~/Castle/Cellar
[mission 31] $ kill 175310 175311 175312

~/Castle/Cellar
[mission 31] $ ps
  PID TTY          TIME CMD
 49087 pts/0    00:00:00 zsh
 49122 pts/0    00:00:00 bash
 49162 pts/0    00:00:02 bash
175283 pts/0    00:00:00 nice_fairy
175284 pts/0    00:00:00 mischievous_imp
175290 pts/0    00:00:00 spell
175291 pts/0    00:00:00 spell
175292 pts/0    00:00:00 spell
175293 pts/0    00:00:00 tail
175313 pts/0    00:00:00 tail
210367 pts/0    00:00:00 sleep
210368 pts/0    00:00:00 sleep
210435 pts/0    00:00:00 sleep
210440 pts/0    00:00:00 sleep
210509 pts/0    00:00:00 sleep
210510 pts/0    00:00:00 ps

~/Castle/Cellar
[mission 31] $ rm *coal*

~/Castle/Cellar
[mission 31] $ gc

Congratulations, mission 31 has been successfully completed!
```

La missione 32 chiede semplicemente di eseguire dei calcoli matematici.

```
~/Castle/Cellar
[mission 32] $ gg

Mission goal

To get better in the magical art, one needs to know mental math.

Get ready, because Merlin is about to test your precision with sums.

Run the command ``gsh check`` to start.

~/Castle/Cellar
[mission 32] $ gc
67 + 38 = ?? 105
18 + 84 = ?? 102
63 + 71 = ?? 134
50 + 93 = ?? 143
100 + 19 = ?? 119

Congratulations, mission 32 has been successfully completed!
```

La missione 33 richiede di risolvere rapidamente una lunga serie di moltiplicazioni. Tuttavia, per via del limite di tempo, non è possibile inserire le operazioni manualmente. La soluzione consiste nell'utilizzare il reindirizzamento dell'input in modo che il comando legga automaticamente le operazioni presenti nel file Mathematics_101.

Il comando **gc (gsh check) < Mathematics_101** sostituisce l'input da tastiera con il contenuto del file, permettendo a gc di elaborare tutte le moltiplicazioni in modo immediato.

```
Mission goal

To get better in the magical art, one needs to know mental math.

Get ready, because Merlin is about to test your speed with products.

Run the command ``gsh check`` to start.

Remark

There now is a time constraint.

Hint

The library is rumored to contain some mathematics books and hidden volumes.

Useful commands

COMMAND < FILE
Replace the command's standard input by a file.
Instead of reading lines from the keyboard device, the command will read lines from the file.

~/Castle/Main building/Library
[mission 33] $ gc < Mathematics_101
96 + 40 = ?? 59 * 60 = ?? 78 * 52 = ?? 55 * 16 = ?? 51 * 65 = ?? 23 * 31 = ?? 81 * 97 = ?? 31 * 21 = ?? 72 * 15 = ?? 90 * 55 = ?? 73 * 20 = ?? 96 * 55 = ?? 18 * 45 = ?? 47 * 55 = ?? 49 * 39 = ?? 20 * 30 = ?? 40
+ 18 = ?? 53 * 79 = ?? 64 * 81 = ?? 12 * 24 = ?? 50 * 78 = ?? 15 * 60 = ?? 53 * 63 = ?? 18 * 14 = ?? 56 * 92 = ?? 47 * 74 = ?? 83 * 32 = ?? 74 * 17 = ?? 74 * 86 = ?? 54 * 96 = ?? 21 * 62 = ?? 12 * 64 = ?? 43 *
60 = ?? 33 * 41 = ?? 58 * 13 = ?? 67 * 36 = ?? 26 * 36 = ?? 82 * 94 = ?? 29 * 12 = ?? 16 * 93 = ?? 65 * 69 = ?? 51 * 88 = ?? 81 * 18 = ?? 61 * 32 = ?? 69 * 94 = ?? 82 * 46 = ?? 1 * 73 = ?? 89 * 23 = ?? 63 * 89
= ?? 74 * 14 = ?? 14 * 52 = ?? 18 * 58 = ?? 72 * 66 = ?? 64 * 40 = ?? 33 * 87 = ?? 27 * 80 = ?? 5 * 5 = ?? 77 * 80 = ?? 80 * 58 = ?? 91 * 100 = ?? 46 * 32 = ?? 79 * 20 = ?? 96 * 92 = ?? 64 * 38 = ?? 48 * 41 =
?? 8 * 51 = ?? 35 * 21 = ?? 25 * 57 = ?? 88 * 28 = ?? 54 * 10 = ?? 66 * 11 = ?? 60 * 69 = ?? 44 * 79 = ?? 41 * 91 = ?? 58 * 20 = ?? 38 * 10 = ?? 59 * 99 = ?? 2 * 34 = ?? 75 * 92 = ?? 34 * 53 = ?? 49 * 56 = ?? 2
+ 100 = ?? 84 * 22 = ?? 46 * 5 = ?? 43 * 65 = ?? 17 * 35 = ?? 49 * 56 = ?? 11 * 86 = ?? 77 * 48 = ?? 99 * 15 = ?? 27 * 47 = ?? 36 * 56 = ?? 34 * 85 = ?? 95 * 38 = ?? 87 * 2 = ?? 4 * 27 = ?? 93 * 22 = ?? 11 *
89 = ?? 12 * 64 = ?? 18 * 72 = ??

congratulations, mission 33 has been successfully completed!
```

La missione 34 richiede di creare un elenco contenente unicamente i libri di incantesimi presenti nell'ufficio di Merlino e di salvarlo nel file inventory.txt, situato nella cartella Drawer. Per identificare correttamente solo gli spell books è stato utilizzato `*grimoire*`, che permette di selezionare tutti i file il cui nome contiene la parola "grimoire", escludendo gli altri.

Il reindirizzamento `>` scrive l'output del comando nel file indicato, generando così l'inventario richiesto. Infine, il comando `gc` ha verificato e confermato la corretta esecuzione della missione.

```

  \_/_/  ||  Mission goal
  ||
  ||  Merlin's old spell books are kept in his office, in the library. You need to save a list of all those spell books (and nothing else) in a file called "inventory.txt", in the
  ||  drawer...
  ||
  ||  Useful commands
  ||
  ||  COMMAND > FILE
  ||  Send the command's output to a file instead of printing it on the screen.
  ||
  ||  less FILE
  ||  display the content of a file, one page at a time
  ||
  ||  Important keybindings are
  ||  q      quit
  ||  Space  scroll down one page
  ||  / STRING search for a string
  ||  n      go to the next occurrence of the
  ||         search string
  ||
  ||  ls FILE1 ... FILEn
  ||  Show the list of files given as arguments.
  ||  This is particularly useful if you use shell patterns with wildcards.
  ||
  { /o\ }
  \_/_/  }

~/Castle/Main_building/Library/Merlin_s_office
[mission 34] $ ls *grimoire* > Drawer/inventory.txt

~/Castle/Main_building/Library/Merlin_s_office
[mission 34] $ gc

Congratulations, mission 34 has been successfully completed!
```



La missione 35 richiede di identificare tutti i libri di alchimia che contengono la stringa “gsh”, indipendentemente dalle maiuscole/minuscole, e di mostrarne esclusivamente il nome del file. Alcuni libri risultano bloccati e generano errori quando si tenta di leggerli, pertanto questi messaggi devono essere soppressi.

Per soddisfare i requisiti, è stato utilizzato il comando: **grep "gsh" -i -l *grimoire* 2>/dev/null**

Le opzioni **-i** e **-l** permettono rispettivamente una ricerca case-insensitive e la stampa dei soli nomi dei file. ***grimoire*** seleziona tutti i libri di alchimia, mentre **2>/dev/null** elimina gli errori prodotti dai file bloccati. L'output finale presenta solo l'elenco corretto dei libri contenenti il composto alchemico richiesto.

```

Mission goal
-----
Merlin's old alchemy books are kept in his office, in the library. You need to output a list of all the books containing the alchemical compound 'gsh' (for "Glutathione"). Beware, it can be spelled with a mix of letters in upper and lowercase.

Some of those books are locked, and you are not allowed to consult them. Ignore them.

Remark
-----
Your last command should display the list of books mentioning 'gsh', and nothing else. In particular, you should not show the chemical compounds themselves and no error message should appear.

Useful commands
-----
grep STRING FILE1 ... FILEN
  Search for a string inside a file and print the corresponding lines.
  Some interesting options are
  -l : do not print the lines, but only the filenames containing the string
  -i : do not differentiate between uppercase and lowercase

COMMAND > FILE
  Send messages (stdout) to the file.
  (The file is overwritten.)

COMMAND 2> FILE
  Send error messages (stderr) to the file.
  (The file is overwritten.)

/dev/null
  "Virtual" file that is universally empty: anything sent to it disappears.

Explanations
-----
Every command can output messages to the screen in two ways:
- standard printing ("stdout")

```

```

~/Castle/Main_building/Library/Merlin's_office
[mission 35] $ grep "gsh" -i -l *grimoire* 2> /dev/null
grimoire_aKeNb1PofQ
grimoire_bWsqGhGUUFsgjdUXa0kkEhyBtvwod
grimoire_CpueNoXrKeVxmMnjhiFYLHXfIENqc
grimoire_eHGNfKnBYSGbVbuznArFdw
grimoire_eiNPrUbStyxAndBXv
grimoire_EyIyAwBBZPafzQzxCnbEPwd
grimoire_EyPboKIeevKpEQiOreHMnhxFbTNzK
grimoire_FTjEJGnMhOVorRkEGcXgEqpnDPqXQ
grimoire_gF0pGYNhSSddFMjv
grimoire_GLzCvCtmobHDLQj
grimoire_isRMLkDNvgJtj0cHCzyVbcSBees0
grimoire_iVxJCziwStfntcbbkKDIOkVZiGz
grimoire_JkUzQmZbizjdVXNLXdNgqCZLGzxsBG
grimoire_KZBswuMbdMFbeOGwvtOLvt
grimoire_lVJcHCqbAIEOXzCpztaiTJMyu
grimoire_nISzmSEDRYMHxMXJfbqcGupeOco
grimoire_nnwAkMmfJOaRdvYxLuZyqjFZl
grimoire_qVwvCpkjZhimwsU
grimoire_RUOHlaYssQh
grimoire_TtpRpJYoLxmeXVxmTaLPSqL
grimoire_udJjHqsZQ
grimoire_UrZcUxalrpfIQGdSEdaDKKZXVzCgs
grimoire_UwuQuYlGTjaGSjVuDJFhVLDmKIqEiKc
grimoire_xjqGhhVfvrjOWieqFFEXjgTtbLxI

~/Castle/Main_building/Library/Merlin's_office
[mission 35] $ gc

Congratulations, mission 35 has been successfully completed!

```

La missione 36 richiede di estrarre una secret key dal programma Merlin, il cui output è contaminato da numerosi messaggi di errore (stderr). Il primo passo è stato individuare l'eseguibile tramite **find -iname merlin**. Una volta raggiunta la directory corretta, si è eseguito il programma filtrando i messaggi di errore, reindirizzando lo stderr verso /dev/null: **./Merlin 2> /dev/null**.

In questo modo è stato possibile isolare la secret key. Successivamente ho salvato l'output valido in un file: **./Merlin 2> key.txt**

La missione richiede di fornire la chiave nuovamente al programma utilizzando un reindirizzamento dell'input. Si è quindi passato il contenuto del file tramite stdin: **gc < key.txt**

```
(0) >>>
Mission goal
Merlin has turned crazy... He paces around the observatory tower and mumbles incoherently.
You need to filter out his ramblings to discover the secret key he is the only one to know.
Hint
It is likely that Merlin's message is riddled with errors.
To complete this mission, you have to give the secret key **using a file redirection**.
Useful commands
./FILE
Run an executable file as a program. Needless to say that Merlin is executable...
COMMAND > FILE
Send standard output (stdout) for the command to a file.
(The file is overwritten.)
COMMAND 2> FILE
Send error output (stderr) for the command to a file.
(The file is overwritten.)
COMMAND < FILE
Send the file to the standard input (stdin) of the command.
(0) >>>

[mission 36] $ find -iname *merlin*
./Castle/Observatory/merlin
./Castle/Main_building/Library/Merlin_s_office

[mission 36] $ cd Castle/Observatory
~/Castle/Observatory
[mission 36] $ ls
merlin star_chart

~/Castle/Observatory
[mission 36] $ ./merlin 2> /dev/null
THESECREKEYISONSTDERR

~/Castle/Observatory
[mission 36] $ ./merlin
BZPCzTzWzFwAHAcYfLUBgIOEmSOECoyTePjD0puxTgV0aRwZsweeEeHuqSbISxEqTgouoBRfRzWSaElKfOnonZuuyIPesZEc0QYIUNLLkxQZatYHcOCWj6ZkSwwLOUUTzQNAmjksAnK1JJToFFVRTCrbaWDIAkgOEZChtttEPYRuBKSGmSRcsRcbZjqTqXYvpvddcFVInub
VSKecThDgLB0

~/Castle/Observatory
[mission 36] $ ./merlin 2> key.txt
THESECREKEYISONSTDERR

~/Castle/Observatory
[mission 36] $ ls
key.txt merlin star_chart

~/Castle/Observatory
[mission 36] $ gc < key.txt
What is the secret key?
Congratulations, mission 36 has been successfully completed!
```

La missione 37 richiede di accedere alla directory "Kings_quarter", situata nella *Throne Room*. Dopo aver individuato la posizione con `find -iname *Throne*` si sono verificati i permessi della directory con `ls -l`, scoprendo che "Kings_quarter" aveva permessi `rw-rw-r--`, cioè senza permesso di esecuzione. Per modificare i permessi è stato utilizzato il comando `chmod +x Kings_quarter`.

Dopo aver modificato i permessi, è stato possibile accedervi correttamente con `cd`.

```
chmod [OPTIONS] FILE
  Modify the permissions for the file (or directory).

  Refer to the manual to discover which options are available.

~/Castle/Observatory
[mission 37] $ cd

[mission 37] $ find -iname *Throne*
~/Castle/Main_building/Throne_room

[mission 37] $ cd Castle/Main_building/Throne_room

~/Castle/Main_building/Throne_room
[mission 37] $ ls -l
total 4
drw-rw-r-- 2 kali kali 4096 Nov 13 13:26 Kings_quarter/

~/Castle/Main_building/Throne_room
[mission 37] $ cd Kings_quarter/
bash: cd: Kings_quarter/: Permission denied

~/Castle/Main_building/Throne_room
[mission 37] $ chmod +x Kings_quarter/

~/Castle/Main_building/Throne_room
[mission 37] $ cd Kings_quarter/

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 37] $ gc

Congratulations, mission 37 has been successfully completed!
```

La missione 38 chiede di leggere una lettera segreta. All'interno dei King's quarters era presente un file nascosto chiamato `.secret_note`, ma i suoi permessi impedivano la lettura del contenuto. Dopo aver verificato i permessi tramite `ls -l`, si sono modificati i permessi di lettura con `chmod +r .secret_note`.

Una volta reso leggibile, il file è stato aperto con `cat`.

```
The King probably tried to make this note unreadable!

Useful commands

ls -l
  Lists the files with their important meta-data.

  Access permissions are listed first on each line.

chmod [OPTIONS] FILE
  Modifies the permissions for the file (or directory).

  Refer to the manual to discover which options are available.

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 38] $ ls -l
total 8
-rwxrwxr-x 1 kali kali 11 Nov 15 10:20 note
--w--w---- 1 kali kali 11 Nov 15 10:20 .secret_note

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 38] $ chmod +r .secret_note

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 38] $ cat .secret_note
2705248967

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 38] $ gc
What's the combination to open the King's safe? 2705248967

Congratulations, mission 38 has been successfully completed!
```

La missione 39 chiede di leggere un file (crown) all'interno di una cassaforte nascosta.

Per trovarla si è utilizzato il comando **ls -A** (abbreviato con **la**) e modificato i permessi tramite **chmod** in modo da riuscire ad aprirla. All'interno si è trovato il file "crown" e, una volta cambiato i permessi anche a quest'ultimo, si è proceduto con la lettura tramite **cat**.

E' stato poi spostato verso il "Chest" con **mv** per terminare la missione.

```
chmod [OPTIONS] FILE
  Modify the permissions for the file (or directory).

  Refer to the manual to discover which options are available.

~/Castle/Main_building/Throne_room
[mission 39] $ la -l
total 8
drwxrwxr-x 2 kali kali 4096 Nov 15 10:17 Kings_quarter/
d----- 2 kali kali 4096 Nov 15 10:24 Safe/

~/Castle/Main_building/Throne_room
[mission 39] $ chmod +rwx Safe

~/Castle/Main_building/Throne_room
[mission 39] $ cd Safe

~/Castle/Main_building/Throne_room/Safe
[mission 39] $ chmod +rwx crown

~/Castle/Main_building/Throne_room/Safe
[mission 39] $ cat crown
(^.^+.^.^)
\o*o*/
{781_}

~/Castle/Main_building/Throne_room/Safe
[mission 39] $ mv crown ~/Forest/Hut/Chest/

~/Castle/Main_building/Throne_room/Safe
[mission 39] $ gc
What are the 3 digits inscribed on the base of the crown? 781

Congratulations, mission 39 has been successfully completed!
```


La missione 42 chiede di determinare quanto denaro il “King” dovesse ancora pagare, analizzando un singolo file contenente diversi acquisti con soli 3 comandi. Per trovare il file corretto, si è filtrato l’elenco dei file presenti nella “Stall” **con `ls | grep "boring_object"`**

Una volta individuato il file, si sono estratte solo le righe riguardanti gli acquisti effettuati, escludendo quelle già pagate. Per farlo, si è utilizzato una pipeline che seleziona le righe contenenti “King” ma rimuove quelle che includono la parola “PAID”: **`cat <file> | grep "King" | grep -v PAID`**.

L’output mostrava i vari oggetti acquistati e il relativo costo. Sommando i valori, risultava che il Re doveva ancora 18 coppers.

```
You are only allowed 3 commands to find the King's debt.
You can always reset the counter with 'gsh reset', but the whole stall and the debts of everyone will be re-generated as well.

Hint
---
When there are no sub-directories, an alternative to ``find . -name '*boring_object*'`` is to use 'ls' and filter the result with grep:
$ ls | grep "boring_object"

Useful commands
-----
grep [OPTIONS] STRING FILE1 ... FILEn
Filter the files lines, keeping only those that contain the given string.
If no file is given, 'grep' uses stdin.

Useful option
-v : only show the lines that **do not contain** the string.

(1)
~/Stall
[mission 42] $ ls | grep -v 'boring object'
06f585246977_s_c_r_o_l_l_06f58524697725f4
(2)
~/Stall
[mission 42] $ cat 06f585246977_s_c_r_o_l_l_06f58524697725f4 | grep "King" | grep -v PAID
the King bought a spade for 5 coppers.
the King bought a knife for 6 coppers.
the King bought an opal for 2 coppers.
the King bought a chicken for 5 coppers.
(3)
~/Stall
[mission 42] $ gc
How much does the king owe? 18
Congratulations, mission 42 has been successfully completed!
```

La missione 43, invece, chiede di contare quanti oggetti sono presenti nella “Stall” che risultano “UNPAID” in un unico comando.

Per filtrare correttamente i file rilevanti, si sono esclusi quelli il cui nome conteneva “boring_object” utilizzando **`grep -v 'boring_object'`**.

Successivamente, tramite xargs, si è passata la lista filtrata a grep, escludendo tutte le righe che contenevano la parola “PAID” **`ls | grep -v 'boring_object' -v | xargs grep -v "PAID"`**

Infine, per ottenere il conteggio totale degli oggetti ancora da pagare, si è utilizzato wc: **`ls | grep -v 'boring_object' -v | xargs grep -v "PAID" | wc`**

Il primo valore restituito rappresentava il numero di righe, cioè **38 oggetti non pagati**.

```
(0)
~/Stall
[mission 43] $ ls | grep -v 'boring_object' -v | xargs grep -v "PAID" | wc
38      337      1854
(1)
~/Stall
[mission 43] $ gc
How many unpaid items are there? 38
Congratulations, mission 43 has been successfully completed!
```

Nella missione 44, nel cassetto dell'ufficio di Merlino era presente un file chiamato `secret_message`, il cui contenuto risultava cifrato tramite una semplice sostituzione alfabetica. Per decodificare il messaggio si è utilizzato il comando `tr`, applicando una rotazione dell'alfabeto: `cat secret_message | tr "a-z" "o-za-n"`

Il comando `tr` esegue una sostituzione carattere per carattere, mappando ogni lettera del primo insieme (a-z) sulla lettera corrispondente del secondo insieme (o-za-n).

Quest'ultimo insieme rappresenta l'alfabeto ruotato di 14 posizioni (ROT14), ottenuto concatenando le sequenze o-z e a-n.

In pratica, ogni lettera del messaggio cifrato viene sostituita con quella che si trova 14 posizioni più avanti nell'alfabeto, consentendo di decodificare correttamente il testo.

Il testo risultante conteneva le ultime volontà di Merlino e rivelava la parola chiave necessaria per far riapparire il suo baule. La chiave indicata nel messaggio era: `qdqn`

```
~/Castle/Main_building/Library/Merlin_s_office/Drawer
[mission 44] $ cat secret_message | tr "a-z" "o-za-n"
here is my will:
you will get my chest, and everything it contains.
this chest is in the cellar, and the word to make
it re-appear is: qdqn
merlin the enchanter

~/Castle/Main_building/Library/Merlin_s_office/Drawer
[mission 44] $ gc
What's the key that will make Merlin's chest to appear?
qdqn

Congratulations, mission 44 has been successfully completed!
```



Una volta concluse tutte le missioni si è avuta conferma tramite questa schermata.

```
[ progress was saved in /home/kali/gameshell-save.sh ]

  \ : /      \ : /      \ : /      \ : /      \ : /
- . : . -    - . : . -    - . : . -    - . : . -    - . : . -
  \ : /      \ : /      \ : /      \ : /      \ : /

  \ : /      \ : /      \ : /      \ : /      \ : /
- . : . -    - . : . -    - . : . -    - . : . -    - . : . -
  \ : /      \ : /      \ : /      \ : /      \ : /

CONGRATULATIONS!

You have finished all the missions.

  \ : /      \ : /      \ : /      \ : /      \ : /
- . : . -    - . : . -    - . : . -    - . : . -    - . : . -
  \ : /      \ : /      \ : /      \ : /      \ : /

  \ : /      \ : /      \ : /      \ : /      \ : /
- . : . -    - . : . -    - . : . -    - . : . -    - . : . -
  \ : /      \ : /      \ : /      \ : /      \ : /
```

```
~/Castle/Main_building/Library/Merlin_s_office/Drawer
[mission 45] $ gsh index
1  basic/01_cd_tower (completed)
2  basic/02_cd.._cellar (completed)
3  basic/03_cd_HOME_throne (completed)
4  basic/04_mkdir_chest (completed)
5  basic/05_rm_spiders_cellar (completed)
6  basic/06_mv_coins_garden (completed)
7  basic/07_mv_hidden_coins_garden (completed)
8  basic/08_rm_wildcard_spiders_cellar (completed)
9  basic/09_rm_wildcard_hidden_spiders_cellar (completed)
10 basic/10_cp_standard_great_hall (completed)
11 basic/11_cp_wildcards_tapestries_great_hall (completed)
12 basic/12_cp_ls_mtime_paintings_tower (completed)
13 misc/01_cal_nostradamus [optional] (completed)
14 intermediate/01_alias_la (completed)
15 misc/02_nano_journal (completed)
16 intermediate/02_alias_journal (completed)
17 intermediate/03_tab_spider_lair (completed)
18 intermediate/04_bg_xeyes [optional] (completed)
19 intermediate/05_background (completed)
20 intermediate/06_control-C (completed)
21 finding_files_maze/01_ls_cd (completed)
22 finding_files_maze/02_tree (completed)
23 finding_files_maze/03_find_1 (completed)
24 pipe_intro_book_of_potions/01_head (completed)
25 pipe_intro_book_of_potions/02_tail (completed)
26 pipe_intro_book_of_potions/03_cat (completed)
27 pipe_intro_book_of_potions/04_pipe (completed)
28 pipe_intro_book_of_potions/05_pipe_head_tail (completed)
29 processes/01_ps_kill (completed)
30 processes/02_ps_kill_signal (completed)
31 processes/03_pstree_kill (completed)
32 stdin_stdout_stderr/01_stdin_additions (completed)
33 stdin_stdout_stderr/02_stdin_redirection_multiplications (completed)
34 stdin_stdout_stderr/03_stdout_redirection_inventory (completed)
35 stdin_stdout_stderr/04_stderr_dev-null_grimoires (completed)
36 stdin_stdout_stderr/05_stdout_stderr_redirection_merlin (completed)
37 permissions/01_chmod_x_dir_king_quarter (completed)
38 permissions/02_chmod_r_file_king_quarter (completed)
39 permissions/03_chmod_rw_file_dir_throne_room (completed)
40 finding_files_maze/04_find_2 (completed)
41 finding_files_maze/05_find_xargs_grep (completed)
42 pipes_merchant_stall/01_pipe_1 (completed)
43 pipes_merchant_stall/02_pipe_2 (completed)
44 misc/03_tr_caesar_shift (completed)
→ 45 FINAL_MISSION
```

Conclusione

L'attività svolta ha permesso di approfondire in modo pratico e strutturato l'utilizzo della shell Linux attraverso l'impiego del gioco GameShell, rivelatosi uno strumento efficace per l'apprendimento progressivo dei comandi fondamentali e delle dinamiche di navigazione del filesystem.

Il completamento delle missioni ha consentito di consolidare competenze operative quali la gestione di file e directory, la manipolazione dei permessi, l'uso di pipeline e redirezioni, la ricerca di contenuti tramite comandi avanzati e l'interazione con processi e flussi di input/output.

Il percorso ha inoltre favorito lo sviluppo di un approccio metodico alla risoluzione dei problemi, grazie alla lettura attenta delle tracce, alla scomposizione delle missioni in sotto-obiettivi e alla verifica continua mediante gli strumenti forniti dalla piattaforma.

La documentazione prodotta rende il processo replicabile e costituisce una guida di riferimento utile sia per lo studio individuale sia come base per attività didattiche future.

In conclusione, l'esperienza con GameShell ha contribuito in modo significativo all'acquisizione di competenze essenziali nell'ambito della gestione di sistemi Linux, offrendo un contesto pratico, interattivo e adatto allo sviluppo di abilità operative spendibili in ambito sistemistico, cybersecurity e amministrazione di sistemi.

Appendice

Comandi principali utilizzati

Durante l'esecuzione delle missioni sono stati utilizzati numerosi comandi fondamentali della shell Linux. Tra i più significativi si evidenziano:

- **cd, ls, pwd, tree** – navigazione del filesystem
- **cp, mv, mkdir, rm** – gestione di file e directory
- **chmod** – modifica dei permessi su file e cartelle
- **cat, head, tail, less** – lettura e analisi dei file
- **grep, find, xargs** – ricerca avanzata di file e contenuti
- **ps, pstree, kill** – gestione dei processi
- **echo, tr, wc** – manipolazione di stringhe e flussi di dati
- redirezioni **>, <** e pipe **|** per la gestione dell'input/output
- comandi nativi del gioco: **gsh goal, gsh check, gsh index**.

Esempi di procedure ricorrenti

Le missioni presentano schemi risolutivi ricorrenti, tra cui:

1. Lettura dell'obiettivo tramite **gsh goal**.
2. Analisi dell'ambiente di lavoro con **ls**, **tree**, **cat**, **find**.
3. Esecuzione delle operazioni richieste (spostamenti, copie, cambi permessi, ricerche, pipe, filtraggi)
4. Verifica del completamento tramite **gsh check**

Evidenze finali

Al termine dell'esercitazione:

- **tutte le missioni previste sono state completate con successo**, inclusa la missione finale;
- gli output di completamento sono stati acquisiti e riportati nella sezione dedicata del report;
- l'elenco completo delle missioni svolte è consultabile tramite il comando **gsh index**, che conferma il completamento del percorso.

Venezia, 13/11/2025

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